INCH-POUND

A-A-54988
09 December 1994

COMMERCIAL ITEM DESCRIPTION

RESIN, ACRYLIC, DENTAL, Fibered

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification U-R-00179A for Type II which is cancelled.

This Commercial Item Description covers a fibered acrylic base denture resin formed by cold-cure (autopolymerization) suitable for denture repair.

Salient characteristics:

General. The resin shall be in accordance with ANSI/ADA Specification No. 13 unless otherwise specified. The resin shall be an acrylic base denture resin formed by autopolymerization suitable for use in denture repair. The resin shall be supplied in a package containing a bottle of powder, 2 bottles of liquid, and an empty, plastic squeeze-type powder dispenser. The package shall contain a minimum of one pound (454 grams) of powder and 12 fluid ounces (360 milliliters) of liquid.

Material.

Liquid. The liquid shall be a methyl methacrylate monomer with a suitable storage polymerization inhibitor. The liquid shall be clear, colorless, and substantially free from undissolved or particulate matter when examined, without accessory magnification, against a black background and against a white background. Illumination shall be from a light which, at a point 10 inches from its source, provides an intensity of illumination of not less than 200 and not more than 350 foot candles. The liquid shall be a cross-link type monomer, which produces a cross-linked cured resin on polymerization with the powder, when processed in accordance with manufacturer's instructions.

Color. There shall be no visible discoloration of the liquid when tested as specified below.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Thermal stability. The liquid shall show no thickening when heated in the dark in a closed container at 60° ± 2°C for 24 hours.

Powder. The powder shall be composed of polyacrylic acid esters and poly (substituted) acrylic acid esters, or copolymers of either, or a mixture of polymers and copolymers. The powder shall consist of light pink, internally pigmented, spherical particles, chemically constituted to exhibit high viscosity, when plasticized, for form retention during manipulation. Cellulose acetate fibers shall be distributed among the particles. The fibers shall be red in color, 7 to 9 denier, 3/32 ± 1/64 inch in length, and 0.15 to 0.20 percent by weight. The powder shall be free of extraneous material, such as dirt and lint.

Dispenser. The dispenser shall be a 4-ounce size, translucent polyethylene, squeeze-type bottle with a screw-type polyethylene cap. The top of the cap shall be conical in shape, truncated at the apex to form an opening 1/8 ± 1/32 inch in diameter. The nozzle orifice shall permit passage of intermingled fibers and particles during use. The orifice, when not in use, shall be closed with a snap-on, snap-off polyethylene cap.

Properties of uncured resin polymer. The uncured resin polymer, when prepared at 23.0° ± 2.0°C and as directed by the instructions accompanying the package shall be considered ready for packing when plasticity is such that the polymer separates cleanly from the walls of the glass mixing jar. This condition shall be attained in 3-1/2 minutes or less from the start of the mixing.

The resin polymer shall be capable of being intruded into a minimum of two holes of the die to a depth of 0.5 millimeter (mm) minimum when tested in accordance with the packing plasticity test of ANSI/ADA Specification No. 12 (Denture Base Polymers) for Type II autopolymerized polymers. The resin specimen used for the die intrusion test shall have a smooth, uniform surface, free of granulation and mottling during and after gelation.

Properties of cured polymer. The resin polymer, when processed according to instructions furnished by the manufacturer, shall yield a satisfactory denture repair.

Gloss. The resin polymer shall have a glossy surface and shall show no surface defects, when cured in a gypsum mold lined with tin foil.

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Color reflectance. The light pink color of the acrylic repair material shall conform to color reflectance values of 17.0 to 20.0 for "L", 19.0 to 22.0 for "A", and 7.5 to 9.0 for "B" using a Gardner Model AC-1 Color Difference Meter.

Color stability. The color change of the cured polymer shall be no greater than 2.0 delta E units when tested in accordance with ANSI/ADA Specification No. 12.

Porosity. The specimens of the transverse deflection test shall show no bubbles or voids or other defects.

Manufacturer's instructions. Instructions shall accompany each package of resin and shall include a method for repairing the resin polymer.

Certification. The resin shall be certified by the American Dental Association. At time of award of the contract, the manufacturer shall furnish, upon request, a certificate and data showing that the brand of material to be furnished is the same in quality and brand as that appearing on the list of certified materials. Compliance with this certification requirement does not absolve the manufacturer from complete compliance with the other requirements of this specification.

Immediate containers. The powder shall be packaged in a wide-mouth glass bottle of suitable size and design. The liquid shall be packaged in narrow-mouth, light-resistant glass bottles of suitable size and design.

Workmanship. The resin powder and liquid shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing one pound of acrylic resin powder in a bottle, 12 fluid ounces of liquid monomer in 2 bottles, a powder dispenser, and instructions for use, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one package of acrylic resin powder and liquid, with dispenser and instructions, as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 2.5.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one package of acrylic resin powder and liquid Sampling shall be inspection level S-2 with an AQL of 1.0.

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Discoloration test. Discoloration of the liquid monomer shall be determined by placing 10 milliliters (ml) of the liquid into a clear glass test tube, having an inside diameter of approximately 1 cm, and by exposing this sample to bright diffuse daylight for 24 hours. This sample shall be compared with 10 ml of the unexposed liquid placed in a similar test tube. The tubes shall be viewed perpendicularly to their long axis when backed by a white background.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition

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Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-935-4013	RESIN, ACRYLIC, DENTAL, Fibered, Denture Base Repair, Light Pink, 1 lb.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS
GSA-FSS-FCGC

Project No. 6520-2581
Location: c:finals/AA-54988/mar

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STANDARDIZATION DOCUMENT

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	A-A-54988	941209

3 DOCUMENT TITLE
RESIN, ACRYLIC, DENTAL, Fibered

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-MAP	(1) Commercial (215) 737-8431	(2) AUTOVON 444-8431
ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756 2340 AUTOVON 289-2340	

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